M1 Garand Rifle

Springfield Armory / Winchester / H&R; / BMR • U.S. Rifle, Caliber .30, M1 'The Greatest Battle Implement Ever Devised'

Manufacturer:	Springfield Armory / Winchester / H&R; / BMR	**Category:**	Rifles & Carbines
Action Type:	Long-Stroke Gas Piston Semi-Automatic	**Era / Service:**	WWII & Korean War Legend (1936–1957)
Calibers:	.30-06 Springfield (M2 Ball), 7.62x51mm NATO (.308 Win)	**Covered Models:**	M1 Garand Standard WWII, Post-War M1, M1C / M1D Sniper, Navy 7.62mm Conversion

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Operating Rod Tilt Test • Gas Cylinder Diameter Gauging (0.5320") • Op-Rod Piston Head Gauging (0.5260") • Clip Latch Timing & Ejection • M1 Grease Protocol (Lubriplate 130A)

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
TILT TEST PROTOCOL	Assemble bare receiver, barrel, op-rod, and bolt (no spring). Tilt rifle 45° up and 45° down; operating rod and bolt MUST slide back and forth under own weight without binding
GAS CYLINDER DIAMETER	0.5280" minimum to 0.5320" maximum allowable rejection limit (gauge with cylinder bore plug)
PISTON HEAD DIAMETER	0.5260" minimum allowable wear limit (measured at forward edge of op-rod piston)
OPERATING ROD TAB	0.315" wide by 0.095" thick; worn tab under 0.085" will dismount from receiver track during firing
EN BLOC CLIP EJECTION	Clip must eject with distinct metallic ping when final 8th round is fired; clip latch spring must be fresh

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Stoppage 1: Bent Operating Rod from Commercial High-Power Ammo **Operating rod binds in stock channel; rifle jams out of battery**	• Firing modern commercial heavy-bullet (180gr+) hunting ammunition with slow-burning powder. • Excessive port pressure at muzzle gas port bending the delicate dogleg curve of the op-rod.	**Immediate Action: DO NOT shoot commercial hunting ammo without vented gas plug. Gunsmith Action: Check op-rod on Garand alignment fixture. Straighten op-rod dogleg curves on lead block. Install Schuster or McCann adjustable vented gas cylinder lock screw to vent excess pressure when shooting modern ammunition.**
Stoppage 2: Premature Clip Ejection (Dumping 7 Live Rounds with Clip) **First round fires, and the gun ejects the en-bloc clip and remaining 7 live cartridges**	• Worn clip latch or weak clip latch spring. • Bullet guide accelerator cam ledge worn round. • Worn operating rod catch pin.	**Gunsmith Action: Push out clip latch pin. Replace clip latch spring with new extra-power spring. Inspect clip latch engagement hooks and bullet guide lobes with dial calipers; replace components if rounded beyond USGI specifications.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
M1 Garand Complete Detail Disassembly, Inspection & Tilt Test	Brownells Gunsmithing Bench	Full USGI armorer teardown of the M1 Garand: performing the 45° tilt test, gauging the gas cylinder, inspecting the op-rod tab, and applying Lubriplate 130A.